

Theorem 22.1 let $\{\vec{u}_1, \dots, \vec{u}_m\}$ and $\{\vec{v}_1, \dots, \vec{v}_n\}$ be orthonormal and let $\sigma_1, \dots, \sigma_r$ be positive

Then if $A\vec{v}_i = \begin{cases} \sigma_i \vec{u}_i & i=1, \dots, r \\ \vec{0} & i=r+1, \dots, n \end{cases}$

$$= U \Sigma V^*$$

we have that

i. $\text{range } A = \text{span}(\vec{u}_1, \dots, \vec{u}_r)$

ii. $\text{range } A^* = \text{span}(\vec{v}_1, \dots, \vec{v}_r)$

iii. $\text{null } A = \text{span}(\vec{v}_{r+1}, \dots, \vec{v}_n)$

iv. $\text{null } A^* = \text{span}(\vec{u}_{r+1}, \dots, \vec{u}_m)$

Exercise :

1) let A have SVD of

$$A = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} \sqrt{8} & 0 & 0 \\ 0 & \sqrt{2} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & -1 & 0 \\ 0 & 0 & \sqrt{5} \\ 1 & 2 & 0 \end{bmatrix}$$

$\underbrace{\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix}}_{\text{range } A} \underbrace{\begin{bmatrix} \sqrt{8} & 0 & 0 \\ 0 & \sqrt{2} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}}_{\text{null } A^*} \underbrace{\begin{bmatrix} 2 & -1 & 0 \\ 0 & 0 & \sqrt{5} \\ 1 & 2 & 0 \end{bmatrix}}_{\text{range } A^*} \underbrace{\quad}_{\text{null } A}$

rank $A = 2$

find orthonormal bases for the fundamental

subspaces of A .

$$\text{range } A = \text{span} \left(\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right)$$

$$\text{range } A^T = \text{span} \left(\frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right)$$

$$\text{null } A = \text{span} \left(\frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} \right)$$

$$\text{null } A^T = \text{span} \left(\frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right)$$

Theorem 22.2 let $T \in \mathcal{L}(\mathbb{R}^2, \mathbb{R}^2)$ be invertible and let $S = \{ \vec{x} \in \mathbb{R}^2 : \vec{x}^T \vec{x} = 1 \}$ be the unit circle. Then $T[S]$ is an ellipse.

Proof: let $A = [T]_{\{\vec{e}_i\}}$ and set

$\vec{y} = T(\vec{x}) = A \cdot \vec{x}$. Then

$$1 = \vec{x}^T \vec{x} = (\underbrace{A^{-1} A \vec{x}}_{\vec{y}})^T (\underbrace{A^{-1} A \vec{x}}_{\vec{y}})$$

$$= \vec{y}^T \left(\underbrace{(A^{-1})^T A^{-1}}_{\text{pos. def. as } A^{-1} \text{ is full rank}} \right) \vec{y}$$

pos. def. as A^{-1} is full rank

is an ellipse as $(A^{-1})^T A^{-1}$ is pos. def.

Exercise:

2) Find the image of the unit circle under the linear map

$$T\vec{e}_1 = 2\vec{e}_1 + \vec{e}_2$$

$$T\vec{e}_2 = 3\vec{e}_1 + 4\vec{e}_2$$

Solution: setting $A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$ we have

that $A^{-1} = \frac{1}{5} \begin{bmatrix} 4 & -3 \\ -1 & 2 \end{bmatrix}$ and thus

$$(A^{-1})^T A^{-1} = \frac{1}{25} \begin{bmatrix} 4 & -1 \\ -3 & 2 \end{bmatrix} \begin{bmatrix} 4 & -3 \\ -1 & 2 \end{bmatrix}$$

$$= \frac{1}{25} \begin{bmatrix} 17 & -14 \\ -14 & 13 \end{bmatrix}$$

Now find the e-values and e-vectors

$$\begin{vmatrix} 17-\lambda & -14 \\ -14 & 13 \end{vmatrix} = \lambda^2 - 30\lambda + 25$$

$$\lambda = 15 \pm 10\sqrt{2}$$

$$\lambda = 15 \pm 10\sqrt{2} : \begin{bmatrix} 2 \mp 10\sqrt{2} & -14 \\ -14 & -2 \mp 10\sqrt{2} \end{bmatrix}$$

$$\vec{v} = \begin{bmatrix} 14 \\ 2 \mp 10\sqrt{2} \end{bmatrix}$$

Thus $(A^{-1})^T A^{-1}$ has e-values of

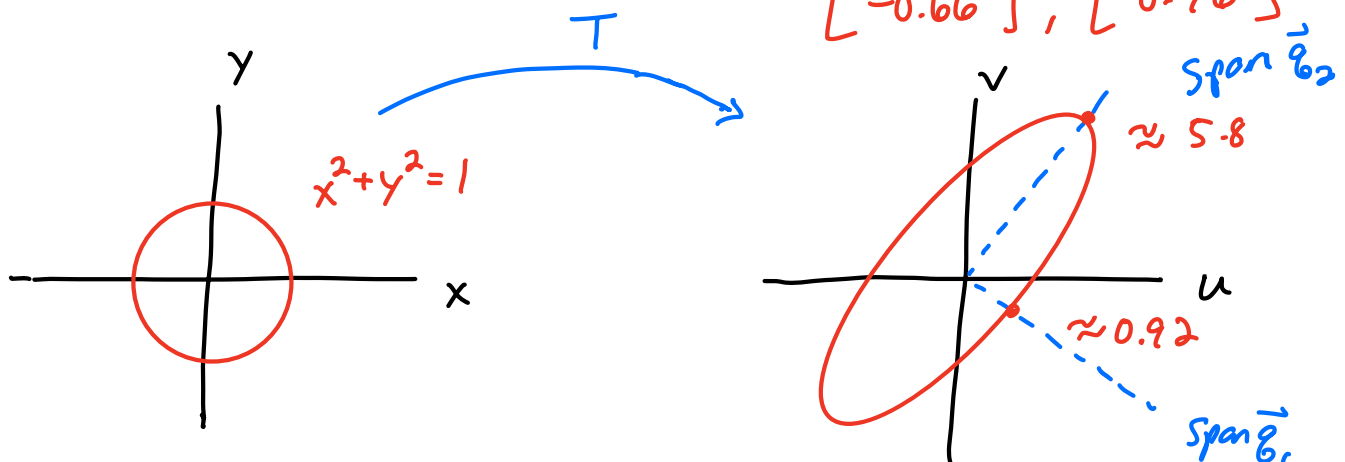
$$\lambda = \frac{15 \pm 10\sqrt{2}}{25} = \frac{3}{5} \pm \frac{2\sqrt{2}}{5} \\ \approx 1.17, 0.03$$

and corresponding e-vectors

$$\vec{e} \approx \begin{bmatrix} 0.76 \\ -0.66 \end{bmatrix}, \begin{bmatrix} 0.66 \\ 0.76 \end{bmatrix}$$

Now the image is an ellipse with half lengths of $\frac{1}{\sqrt{1.17}} \approx 0.92$ and $\frac{1}{\sqrt{0.03}} \approx 5.8$

and principal axes of $\begin{bmatrix} 0.76 \\ -0.66 \end{bmatrix}, \begin{bmatrix} 0.66 \\ 0.76 \end{bmatrix}$



Check: letting $\begin{bmatrix} x \\ y \end{bmatrix} = T^{-1} \begin{bmatrix} u \\ v \end{bmatrix}$

$$= \frac{1}{5} \begin{bmatrix} 4 & -3 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix}$$

$$= \frac{1}{5} \begin{bmatrix} 4u - 3v \\ -u + 2v \end{bmatrix}$$

$$x = \frac{1}{5} (4u - 3v)$$

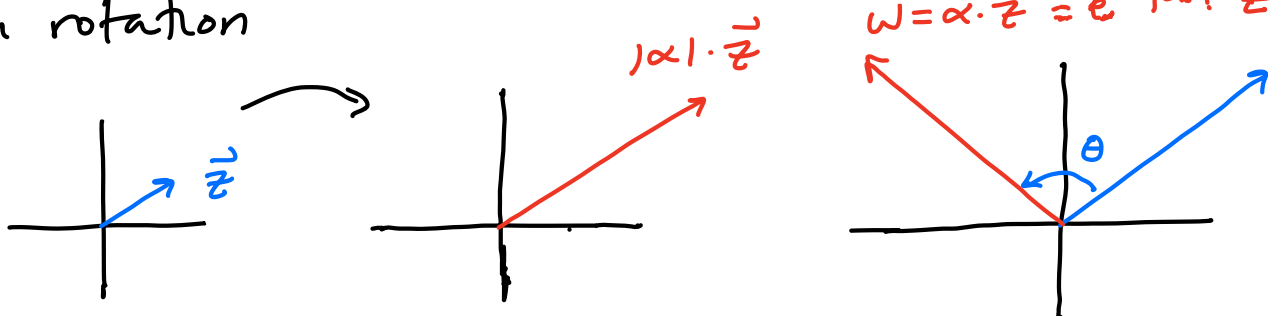
$$y = \frac{1}{5} (-u + 2v)$$

$$x^2 + y^2 = 1 \Rightarrow \left(\frac{4u - 3v}{5} \right)^2 + \left(\frac{-u + 2v}{5} \right)^2 = 1$$

Polar Decomposition

Background: Given complex $\alpha \in \mathbb{C}$, we can write $\alpha = |\alpha| e^{i\theta}$ where $\theta = \arg \alpha = \arctan\left(\frac{\operatorname{Im} \alpha}{\operatorname{Re} \alpha}\right)$

Now the analytic, linear map $w = \alpha \cdot z$ can be realized as the product of a dilation and a rotation



Theorem 22.3 let $A \in \mathbb{C}^{n \times n}$. Then there exists unitary matrix Q such that $A = Q \sqrt{A^* A}$ where $\sqrt{A^* A}$ is the unique positive, semi-definite square root of $A^* A$.

Proof: let the SVD of be given by $A = U \Sigma V^*$

where U and V are unitary. Then

$$A^* A = V \underbrace{\Sigma (U^* U)}_{A^*} \Sigma V^* = V \Sigma^2 V^*$$

and now set $\sqrt{A^* A} = V \Sigma V^*$. Then

$$A = U \Sigma V^* = U V^* V \Sigma V^* = \underbrace{U V^*}_Q \sqrt{A^* A}$$

as $Q = U V^*$ is unitary.

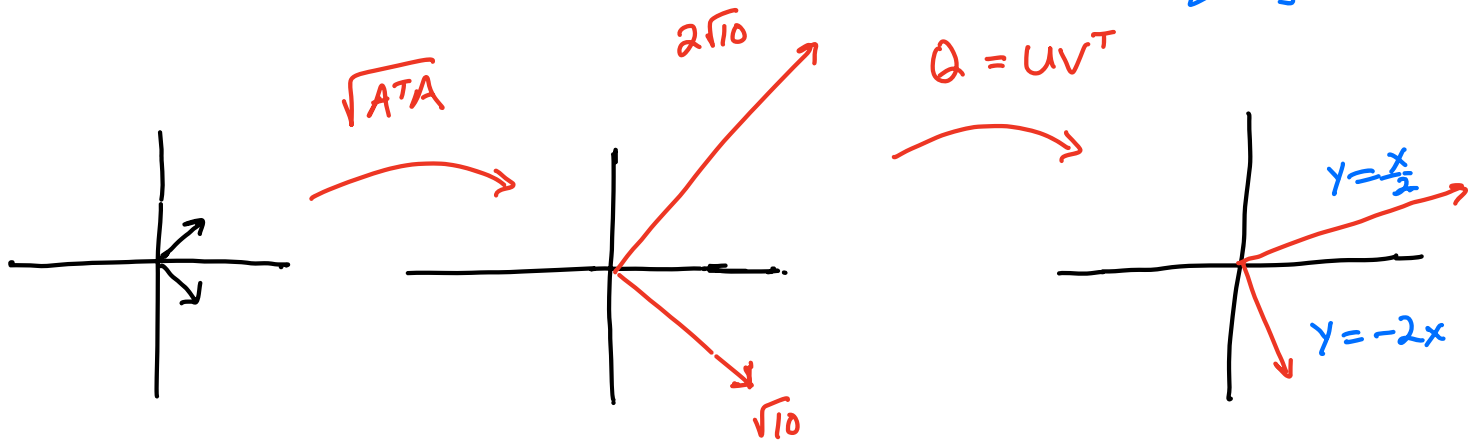
$$(U V^* (U V^*)^* = U V^* V U^* = I)$$

Example: Given $A = \begin{bmatrix} 3 & 5 \\ 4 & 0 \end{bmatrix}$ with SVD

$$A = \underbrace{\frac{1}{\sqrt{5}} \begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix}}_U \underbrace{\begin{bmatrix} \sqrt{40} & 0 \\ 0 & \sqrt{10} \end{bmatrix}}_\Sigma \underbrace{\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}}_{V^T}$$

A has polar decomposition

$$A = \frac{1}{\sqrt{10}} \begin{bmatrix} 1 & 3 \\ 3 & -1 \end{bmatrix} \overset{UV^T}{\frac{\sqrt{10}}{2}} \overset{\sqrt{A^T A}}{\begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}} \rightarrow \text{e-vectors of } \begin{bmatrix} 1 & 1 \\ \pm 1 & \end{bmatrix}$$



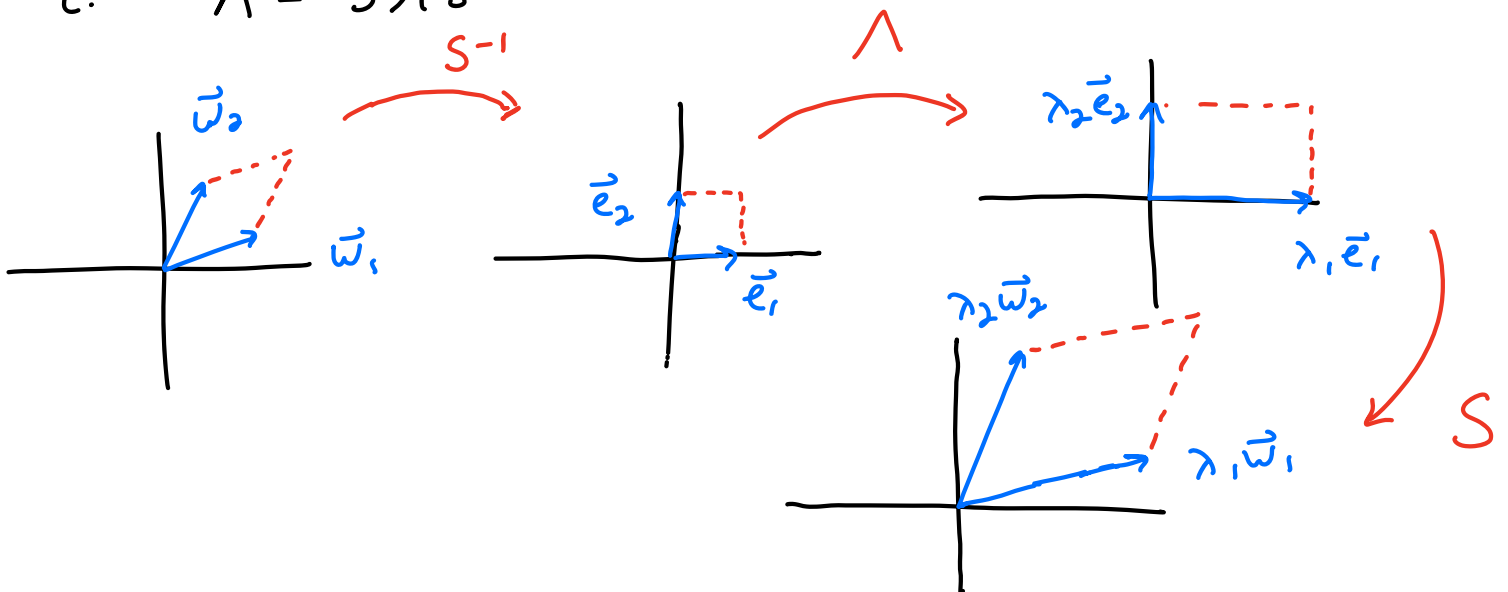
SVD vrs Diagonalization in Pictures

Let $A \in \mathbb{R}^{2 \times 2}$ be such that

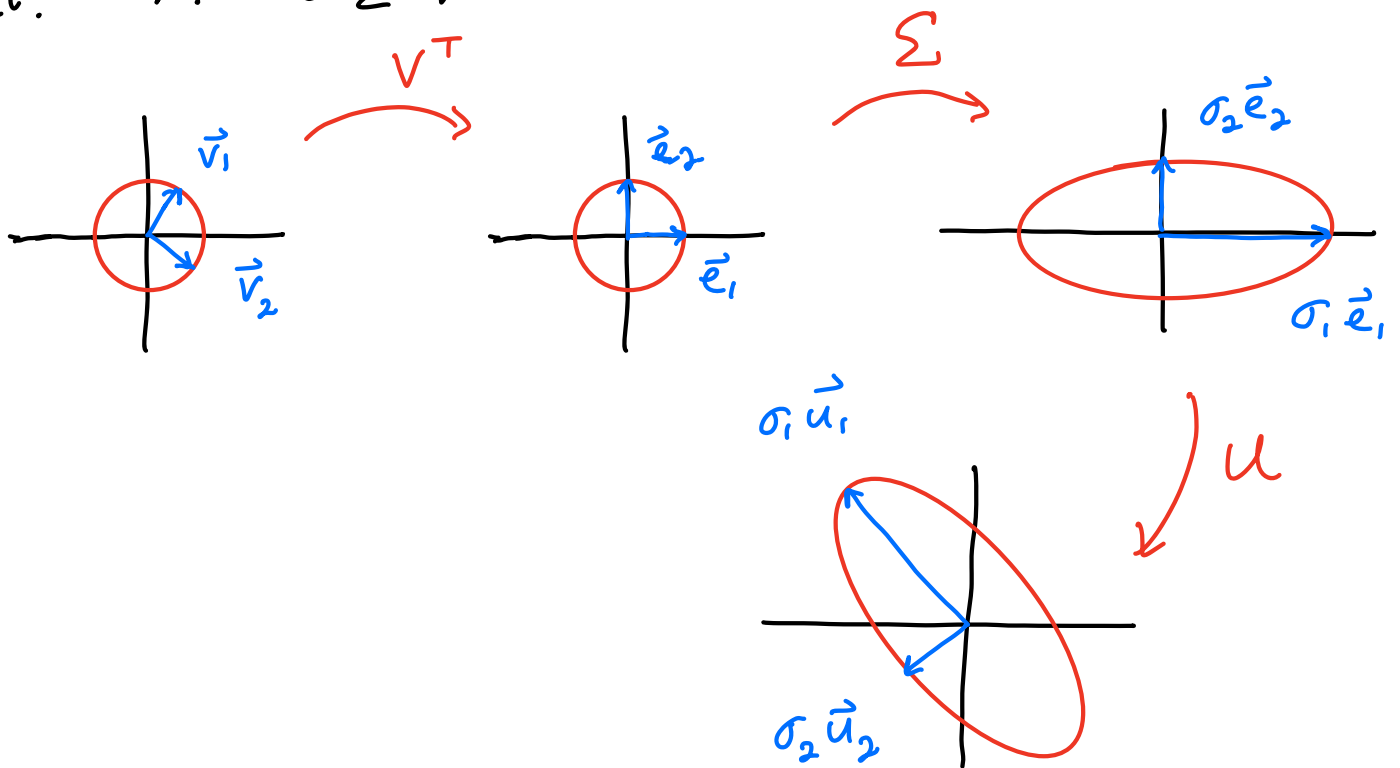
i. $A \vec{w}_i = \lambda_i \vec{w}_i \quad (\lambda_i \neq 0)$

ii. $A \vec{v}_i = \sigma_i \vec{u}_i \quad (\sigma_i \neq 0, \quad \vec{v}_i^T \vec{v}_j = \vec{u}_i^T \vec{u}_j = \delta_{ij})$

i. $A = S \Lambda S^{-1}$



ii. $A = U \Sigma V^T$



Pseudo Inverse (Moore-Penrose Inverse)

Recall that restricting A to $\text{range}(A^*)$ is a bijection and hence is invertible

$$A|_{\text{range } A^*} : \text{range } A^* \rightarrow \text{range } A$$

$$A \vec{v}_i = \begin{cases} \sigma_i \vec{u}_i & i=1, \dots, r \\ \vec{0} & \text{else} \end{cases} \quad \leftarrow$$

$$A|_{\text{range } A^*} : \vec{v}_i \rightarrow \sigma_i \vec{u}_i, \quad i=1, \dots, r$$

$$\sigma_i^{-1} \vec{v}_i \leftarrow \vec{u}_i$$

Def 22.4 let $A = \sum_{i=1}^r \sigma_i u_i v_i^*$ be the reduced SVD of A . Then the pseudo-inverse of A is given

$$A^+ = \sum_{i=1}^r \sigma_i^{-1} v_i u_i^*$$

- $\bar{v}_i$ are e-vectors of A^*A
- $\bar{u}_i$ are e-vectors of AA^*

Properties 22.5

$$i. \quad A^+ \bar{u}_i = \begin{cases} \sigma_i^{-1} \bar{v}_i & i=1, \dots, r \\ \bar{0} & i=r+1, \dots, m \end{cases}$$

ii. If $A = U \Sigma V^*$, then $A^+ = V \Sigma^+ U^*$
(reciprocal transpose)

$$\Sigma = \begin{bmatrix} \sigma_1 & \dots & \sigma_r \\ & & \end{bmatrix}_{m \times n} \quad \Sigma^+ = \begin{bmatrix} \sigma_1^{-1} & & \\ & \dots & \\ & & \sigma_r^{-1} \\ & & & \end{bmatrix}_{n \times m}$$

iii. If A is invertible then $A^+ = A^{-1}$

iv. AA^+ and A^+A are orthogonal projections onto range A and range A^* .

Exercise

3) Given

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 2 & 1 \end{bmatrix}$$

find A^+ and compute AA^+

Finding the SVD of A we see that

$$A^T A = \begin{bmatrix} 2 & 3 & 1 \\ 3 & 6 & 3 \\ 1 & 3 & 2 \end{bmatrix}$$

has e-values satisfying

$$\begin{aligned} 0 &= \begin{vmatrix} 2-\lambda & 3 & 1 \\ 3 & 6-\lambda & 3 \\ 1 & 3 & 2-\lambda \end{vmatrix} \\ &= (2-\lambda)(\lambda^2 - 8\lambda + 3) - 3(3-3\lambda) + (3+\lambda) \\ &= -\lambda^3 + 10\lambda^2 - 9\lambda \\ &= -\lambda(\lambda-9)(\lambda-1) \end{aligned}$$

and thus A has singular values of 3 and 1.

Setting $(A^T A - \lambda_i I) \vec{v}_i = \vec{0}$

$$\lambda_1 = 9: \quad A^T A - 9I = \begin{bmatrix} -7 & 3 & 1 \\ 3 & -3 & 3 \\ 1 & 3 & -7 \end{bmatrix}$$

$$\vec{v}_1 = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

$$\lambda_1 = 1: A^T A - I = \begin{bmatrix} 1 & 3 & 1 \\ 3 & 5 & 3 \\ 1 & 3 & 1 \end{bmatrix}$$

$$\vec{v}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

Finally setting $\vec{u}_i = \frac{1}{\sigma_i} A \vec{v}_i$

$$\begin{aligned} \vec{u}_1 &= \frac{1}{3} A \vec{v}_1 = \frac{1}{3\sqrt{6}} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \\ &= \frac{1}{3\sqrt{6}} \begin{bmatrix} 3 \\ 3 \\ 6 \end{bmatrix} = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} \end{aligned}$$

$$\vec{u}_2 = A \vec{v}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

Now $A^+ = \sum \sigma_i^{-1} \vec{v}_i \vec{u}_i^T$

$$= \frac{1}{3} \cdot \frac{1}{6} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 2 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \begin{bmatrix} 1 & -1 & 0 \end{bmatrix}$$

$$= \frac{1}{18} \begin{bmatrix} 1 & 1 & 2 \\ 2 & 2 & 4 \\ 1 & 1 & 2 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \\ -1 & 1 & 0 \end{bmatrix}$$

$$= \frac{1}{18} \begin{bmatrix} 10 & -8 & 2 \\ 2 & 2 & 4 \\ -8 & 10 & 2 \end{bmatrix}$$

and thus $AA^+ = \frac{1}{18} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 10 & -8 & 2 \\ 2 & 2 & 4 \\ -8 & 10 & 2 \end{bmatrix}$

$$= \frac{1}{18} \begin{bmatrix} 12 & -6 & 6 \\ -6 & 12 & 6 \\ 6 & 6 & 12 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$